Spatial–Channel Attention Transformer With Pseudo Regions for Remote Sensing Image–Text Retrieval

Dongqing Wu, Huihui Li, Yinxuan Hou, Cuili Xu, Gong Cheng, *Member, IEEE*, Lei Guo, and Hang Liu

***Abstract*— Recently, remote sensing image–text retrieval (RSITR) has received significant attention due to its flexible query form and effective management of remote sensing images. However, prior work often relies on compact global features and ignores local features that can reflect salient objects in the images. Moreover, these methods primarily model interactions between features in the spatial domain, which is insufficient for mining the rich semantic information presented in remote sensing images. In this article, we propose a novel spatial–channel attention transformer (SCAT) with pseudo regions to address these issues. Concretely, in order to acquire the fine-grained perception of local objects, we introduce a pseudo region generation (PRG) module that adaptively aggregates grid features with similar semantic information into multiple clusters through a clustering algorithm. These generated cluster centers are able to flexibly and efficiently represent local objects in remote sensing images without relying on sophisticated object detectors. Furthermore, in order to achieve a comprehensive understanding of image semantics information, we carefully construct a novel SCAT. By exploiting spatial and channel attention to explore the dependencies between features at both spatial and channel domains, the proposed SCAT enhances the model's ability to identify both "where to look" and "what it is," thereby obtaining a more powerful representation. In addition, SCAT incorporates two novel designs that alleviate the high overhead caused by attention modeling. Extensive experiments on two benchmark datasets, RSICD and RSITMD, fully demonstrate the effectiveness and superiority of our proposed method.**

***Index Terms*— Attention mechanism, feature clustering, remote sensing image–text retrieval (RSITR), transformer.**

I. Introduction

With the rapid development of remote sensing data acquisition technology, the quantity of remote sensing images has experienced an explosive growth [1], [2], [3]. Effectively managing this large amount of remote sensing data poses a significant challenge before utilizing various methods for image interpretation. Therefore, remote sensing image–text retrieval (RSITR) [4], [5], [6], [7] has emerged as a solution, aiming to use text as a query to retrieve semantically relevant remote sensing images, and vice versa. Due to its flexible query form, RSITR has gained increasing attention recently and has been proven beneficial for many practical applications [8], [9], [10].

Unlike traditional image–text retrieval in natural scenes, RSITR is still in its nascent developmental phase. Concretely, current RSITR methods [4], [5], [11], [12], [13], [14], [15], [16] can be primarily categorized into two types: caption-based methods and embedding-based methods. The former [12], [13] first employs a pretrained remote sensing image captioning model to generate descriptive captions for the images and then performs retrieval by calculating the similarity between the text and the captions. This two-stage retrieval method is essentially a "text-to-text" retrieval process, which not only heavily relies on the quality of captions generated by the captioning model but also inevitably suffers from information loss due to the lack of direct matching between images and texts. To address these problems, the latter [4], [5], [6], [7], [17], [18], [19], [20], [21] map images and texts into a joint high-dimensional semantic space where the similarity between visual and textual embeddings can be directly computed using cosine similarity. For example, Yuan et al. [16] proposed a new network to achieve the fusion of multilevel visual features by constructing a multiscale visual self-attention (SA) model, and proposed a visually guided attention module to obtain better text representation. Mi et al. [17] introduced external knowledge into the retrieval model to obtain more a robust textual representation. Zheng et al. [21] proposed the scale-semantic joint decoupling network to jointly solve scale decoupling and semantic decoupling in a unified model. Compared with the former method, this approach eliminates reliance on the captioning model and significantly reduces information loss, resulting in better retrieval performance.

Despite impressive progress, these methods still face some severe challenges. First, most existing methods only focus on global features and overlook local features that can reflect local objects within the images. Countless studies [22], [23], [24] in image–text retrieval have demonstrated that the task is object-driven, particularly for scene-centric datasets, as shown in Fig. 1(a). In light of this, learning object-level features is crucial to achieving more accurate and interpretable RSITR. To this end, it is a natural thought to use a pretrained object detector to extract multiple salient regions from remote sensing images. However, applying this approach directly to remote sensing images is extremely challenging due to the following

Manuscript received 26 July 2023; revised 24 December 2023 and 29 February 2024; accepted 25 April 2024. Date of publication 30 April 2024; date of current version 9 May 2024. This work was supported by the Key Project of the National Natural Science Foundation of China under Grant 61936007. *(Corresponding author: Huihui Li.)*

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Digital Object Identifier 10.1109/TGRS.2024.3395313

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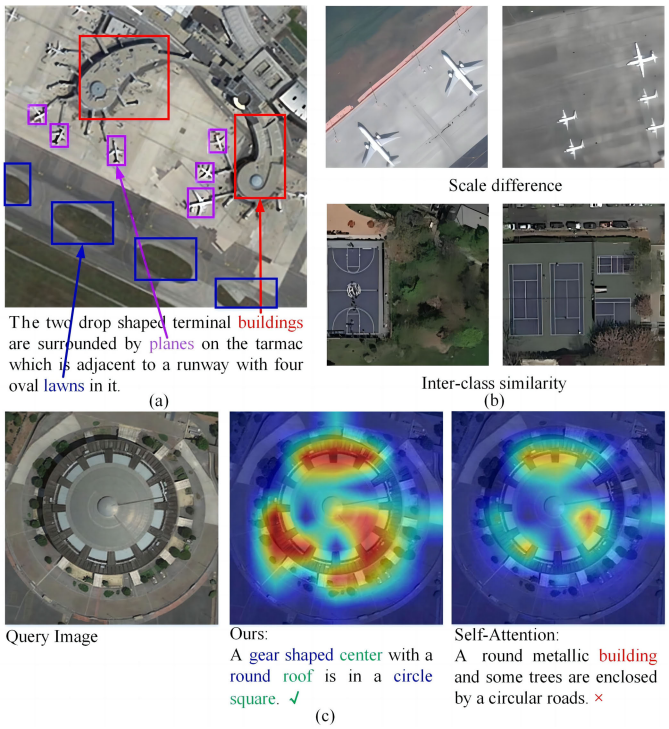


Fig. 1. (a) In scene-centric datasets, identifying local objects in images is crucial for inferring the correlation between images and text. (b) Drawbacks of remote sensing image object detection. (c) Existing SA can capture the dependency between features in the spatial domain (i.e., knowing where to look), but is still insufficient in learning object categories and attributes (i.e., knowing what it is).

reasons. First, the annotation of remote sensing image datasets is costly and time-consuming, which limits the acquisition of training data for object detectors. Second, some inherent characteristics of remote sensing images (i.e., scale difference and interclass similarity) further amplify the challenges and complexities associated with object detection, as shown in Fig. 1(b). Furthermore, the inconsistent optimization objectives of detection and retrieval also prevent these detection-based methods from achieving end-to-end training. This brings us to our first research question: *How can we extract discriminative local features from remote sensing images without using object detectors?*

Second, existing methods lack effective modeling of fine-grained interactions between features, hindering a comprehensive understanding of the content in remote sensing images. They [18], [19], [20], [25] usually rely on the SA mechanism [26] to model spatial dependencies between features, which is insufficient to fully capture semantic information in images. As shown in Fig. 1(c), although the attention map generated by SA can attend to some objects in the image, it cannot accurately identify the attribute and category of the object, resulting in wrong retrieval results. In contrast, our proposed model is able to not only focus on the correct object but also accurately identify the object category and specific attributes (e.g., "center," "roof," and "gear shaped"). This means that the model knowing "where to look" alone is not sufficient, and it also needs to know "what it is" to gain a more comprehensive understanding of the image content. Some methods [27] have demonstrated that different channels of features possess different semantic attributes. Therefore, we can explore interactions between features from a channel perspective to obtain a richer semantic understanding of the image content. Toward this end, an intuitive solution is to directly use the SA mechanism to perform channel-wise interactions. However, the computational complexity of SA is quadratic in the output length, which can lead to significant computational costs when the number of feature channels is large, trapping the model in overfitting. This brings us to our second research question: *How can we comprehensively and effectively model the fine-grained interactions between features in remote sensing images?*

To address these issues, we propose a novel spatial–channel attention transformer (SCAT) with pseudo regions for RSITR. Specifically, we first design a pseudo region generation (PRG) module to overcome the limitations of object detectors and obtain fine-grained perception of local objects in images. In practice, inspired by the NetVLAD operation [28], we utilize a feature clustering algorithm to aggregate low-level grid features into multiple learnable semantic clusters. Each generated cluster describes a semantic theme in the image, with its center referred to as a pseudo region. By aggregating grid features with similar semantic information together, these pseudo regions can effectively represent local objects. Unlike detection-based methods, our PRG module is more flexible, requiring no form of label supervision, and can be trained end-to-end throughout the whole network. Furthermore, we carefully design a novel SCAT to simultaneously model interactions between features in both spatial and channel domains. To capture spatial-wise dependency, the spatial attention block (SAB) in SCAT introduces positional information in two ways: absolute position encoding and relative position weight, which help better model spatial relationships. Meanwhile, we propose a new channel attention block (CAB) to model channel-wise dependency, placing more emphasis on semantically rich channels. To alleviate the high overhead of attention modeling, SCAT introduces two new designs. First, thanks to SCAT's plug-and-play nature, we model the relations among low-level features on pseudo region features, as the number of pseudo regions is much smaller than that of low-level grids. Second, by introducing 1-D convolution, our CAB can avoid the expensive fully connected layers and matrix multiplications present in SA, reducing computational costs. Therefore, our SCAT not only helps the model simultaneously know "where to look" and "what it is" but also addresses the issue of high training overhead, enabling comprehensive and effective modeling of interactions between features. Overall, the main contributions of our article are summarized as follows.

1) We propose a novel PRG module to extract distinctive local features, thereby achieving fine-grained perception of local objects and enhancing the understanding of image content. This approach avoids the need for an additional object detector and can be trained together with the entire network, thus demonstrating stronger adaptability to new data and scenarios.
2) We propose a novel SCAT, which models the interaction between features in remote sensing images from both

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spatial and channel perspectives, thereby obtaining more comprehensive semantic information. The two designs within this module effectively alleviate the high computational cost during attention modeling. Meanwhile, we explore the joint modeling approach for spatial attention and channel attention mechanisms.

3) We conduct extensive experiments on two public datasets, i.e., RSICD [29] and RSITMD [16], to validate the superiority of our model. PRG integrated with SCAT achieves state-of-the-art performance, proving its effectiveness.

II. Related Work

A. Remote Sensing Image–Text Retrieval

With the rapid development of remote sensing data acquisition technology, how to efficiently organize and manage massive amounts of remote sensing data has always been a huge challenge. Therefore, the task of remote sensing retrieval has become a popular direction of research in recent years. Compared with the traditional unimodal retrieval methods [30], [31], [32], cross-modal remote sensing retrieval (i.e., text–image retrieval [19], [20], [21], and audio-image retrieval [33], [34]) has greater versatility because it does not restrict the specific form of the input query. Due to images and text being the two most common modalities, RSITR has attracted widespread attention. Its purpose is to use text (image) as a query to find semantically similar images (text) from the database.

Current RSITR methods can be roughly divided into two categories based on their implementation: caption-based methods and embedding-based methods. The former [12], [13], [35], [36], [37] typically employs an image captioning model to generate captions for remote sensing images and then performs retrieval by computing the similarity between the query text and the captions, which is essentially "text-to-text" retrieval. In this direction, researchers are more focused on image captioning. For example, Shi and Zou [35] constructed an RS image captioning network using fully convolutional networks, which can effectively explore the attributes and interactions of image elements. Lu et al. [29] released a larger dataset, RSICD, to promote further development of image captioning methods. It is noteworthy that the limitations of the two-stage retrieval mode will inevitably lead to information loss, thereby affecting retrieval performance.

The latter [4], [5], [6], [7], [17], [18], [19], [20], [21] directly maps remote sensing images and text into a joint high-dimensional semantic space, where the similarity between images and text can be directly measured by calculating the similarity between features. In this branch, extracting a more powerful and discriminative feature representation is the key goal. To achieve this, Abdullah et al. [11] contributed a new dataset, TextRS, by collecting remote sensing images from four different scenes and annotating each image with five corresponding sentences. They also proposed a bidirectional triplet network composed of convolutional neural network (CNN) and long short-term memory (LSTM) to learn joint embeddings of multimodal data. Rahhal et al. [38] used a larger vision model and a bidirectional LSTM to encode image and text features in an unsupervised learning manner. Considering the multiscale features of remote sensing images, Yuan et al. [16] proposed an asymmetric multimodal feature matching network that generates discriminative visual features through multiscale feature fusion and redundant feature filtering. Meanwhile, they also released a new fine-grained remote sensing image–text match dataset, called RSITMD. Cheng et al. [14] proposed a semantic alignment module to explore the relationship between images and text, thereby producing a more robust visual or textual representation. Furthermore, Yuan et al. [4] proposed a lightweight RSITR model to speed up the retrieval process and improve the retrieval accuracy by knowledge distillation and contrastive learning. Mi et al. [17] introduced external knowledge into the retrieval model to obtain a more robust textual representation. Yu et al. [19] modeled the relationships between objects at different levels in remote sensing images through hyper-graph learning. In a similar vein, Yuan et al. [20] explored the relationships between objects using a graph convolutional network with a denoised representation matrix and enhanced adjacency matrix, and also utilized a dynamic fusion module to integrate global and local information in the image. Zheng et al. [21] proposed the scale-semantic joint decoupling network to jointly solve scale decoupling and semantic decoupling in a unified model.

Despite notable progress, the above methods focus more on global semantic information. Even those methods that consider local information are limited in their applicability in new scenes due to their reliance on object detectors. Unlike them, we propose a flexible and adaptable module to achieve fine-grained perception of local objects in images. This allows for optimization along with the model, freeing it from complex object detectors. In addition, we introduce a more comprehensive feature interaction module to further enhance the understanding of image content and semantics.

B. Transformer Model

As a powerful model for processing sequential data, transformer [26] initially showed great potential in natural language processing tasks. Recently, due to its strong modeling and generalization capabilities, it has been widely applied in computer vision tasks [39], [40], [41], [42], [43], [44], [45], [46]. For example, Bazi et al. [39] were the first to introduce the transformer into the remote sensing image processing field, utilizing vision transformer (ViT) [47] to address remote sensing image classification tasks. Hong et al. [44] employed transformers for hyperspectral image classification, proposing the SpectralFormer. Through groupwise spectral embedding and a cross-layer adaptive fusion scheme, SpectralFormer is capable of learning more valuable information from images. Gao et al. [45] proposed a flexible cross-scale mixing attention transformer (CSMFormer) based on the transformer, achieving satisfactory results in the fusion and classification tasks of multisource remote sensing data. Naturally, many researchers have also applied it to multimodal tasks, such as image captioning [12], [48], [49], [50], visual question answering [51], [52], [53], [54], and visual grounding [10], [55],

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[56]. In the image–text retrieval tasks, Messina et al. [57] proposed a transformer inference network that generates a common concept space by inferring separately on visual and textual modalities. Chen et al. [58] introduced discrete cosine transform (DCT) [59] to the transformer to learn common representation and eliminate irrelevant features from different modalities. Tang et al. [18] proposed the interacting-enhancing feature transformer (IEFT) to fully explore the interactive relationships between images and text. Zhang et al. [60] proposed a framework based on mask-guided relation modeling with entity loss, which utilizes a transformer to facilitate unimodal feature learning on entities and relations.

Our work differs significantly from the above transformer-based image–text retrieval models in terms of the transformer structure. These methods typically capture the spatial dependencies of features through the SA mechanism within transformers, while neglecting channel dependencies. However, directly using SA to model channel relationships leads to excessively high computational complexity. To address this issue, inspired by traditional visual attention, we propose an effective and efficient channel attention model. By integrating the proposed channel attention model into the transformer, our method can simultaneously focus on important spatial and semantic information.

III. Our Method

In this section, we provide a detailed description of the proposed method SCAT-PRG, whose overall architecture is shown in Fig. 2. We begin by introducing the extraction of multilevel visual features in Section III-A, and then provide a detailed description of the proposed PRG module and SCAT module in Sections III-B and III-C, respectively. Moving forward, in Section III-D, we address the processing of textual sentences. Finally, we present the objective function of the model in Section III-E.

A. Visual Representation

Given an input remote sensing image I, we utilize the pretrained ResNet [61] network to extract its visual features, which is consistent with previous methods [4], [16]. Specifically, we flatten the feature map output by the last residual block of ResNet to obtain the global features v^{g} of the image

$$v^{\mathrm{g}} = \mathrm{ResNet}(I, \theta) \tag{1}$$

where θ represents the parameters of the ResNet network, which are fine-tuned based on retrieval tasks.

However, objects in remote sensing images often exhibit multiscale characteristics, whereby the same object appears differently at different scales. For example, a building may appear as small blocks at lower scales but as a complete structure at higher scales. Therefore, using only the feature map output from the last residual block is insufficient for capturing the multiscale characteristics of objects in remote sensing images. In addition, as the number of convolutional layers increases, the network's ability to recognize small targets decreases, which can lead to information loss. Taking into account these factors, we believe that using only global features v^{g} cannot fully and accurately represent the information in remote sensing images, thereby limiting the accuracy of retrieval results. Inspired by other methods [16], [25], we not only extract the global features of remote sensing images but also extract the output features from each block of the ResNet network. The specific extraction process is as follows:

$$v^{\mathrm{g}}, \{v^i\}_{i=2}^5 = \mathrm{ResNet}(I, \theta) \tag{2}$$

$$v^{\mathrm{l}} = \mathrm{Concat}(v^2, \mathrm{Upsample}(v^3)) \tag{3}$$

$$v^{\mathrm{h}} = \mathrm{Concat}(v^4, \mathrm{Upsample}(v^5)) \tag{4}$$

where $\{v^i\}_{i=2}^5$ is the output features of the ith block. Concat() and Upsample(), respectively, denote the concatenation and upsampling operations. v^{l} means low-level features that contain more local detailed information, while v^{h} represents high-level features that focus more on global semantic information. Finally, we use three fully connected layers to map v^{g}, v^{l}, and v^{h} to the d-dimensional embedding space, respectively, to collectively represent the input image.

B. PRG Module

Although we have extracted multiscale features from remote sensing images, these grid-level features do not explicitly represent the object information in the image. In cross-modal image–text retrieval tasks, image–text alignment in the joint embedding space is object-driven. Under this setting, a natural idea is to employ object detection networks to extract multiple salient regions from remote sensing images. Nevertheless, there are several challenges that make object detection for remote sensing images extremely difficult, such as irregular shapes and sizes of objects. To address this issue, inspired by NetVLAD [28] operation, we propose a PRG module that aggregates semantically similar grid features through clustering algorithms. Each aggregated cluster describes a semantic topic in the image, and the center of the cluster, which we call a pseudo region, can effectively represent local objects in the image. As shallow features have more ability to focus on local style and detail information, we choose to perform feature aggregation on v^{l}.

Formally, we first define M learnable vectors as the initial cluster centers $C = \{c_1, c_2, \ldots, c_M\}$. Using dot-product, we can calculate the similarity between each grid feature and these cluster centers. The next step is to assign these grid features to their corresponding clusters. Specifically, given a grid feature v_i^{l}, it is assigned to the jth cluster as follows:

$$\alpha_{i,j} = \frac{\exp\left(v_i^{\mathrm{l}} c_j^T + b_j\right)}{\sum_{m=1}^{M} \exp\left(v_i^{\mathrm{l}} c_m^T + b_m\right)} \tag{5}$$

where b_j is a trainable bias term. By weighted aggregation of all grid features, we can obtain the aggregated feature for each cluster

$$p_m = \mathrm{norm}\left(\sum_{i=1}^{N} \alpha_{i,j}\left(v_i^{\mathrm{l}} - \bar{c}_j\right)\right) \tag{6}$$

where norm represents the $L2$-normalization operation, and N is the number of grid features. $\bar{c}_j$ is another cluster center, called "offset center," represented by a learnable parameter

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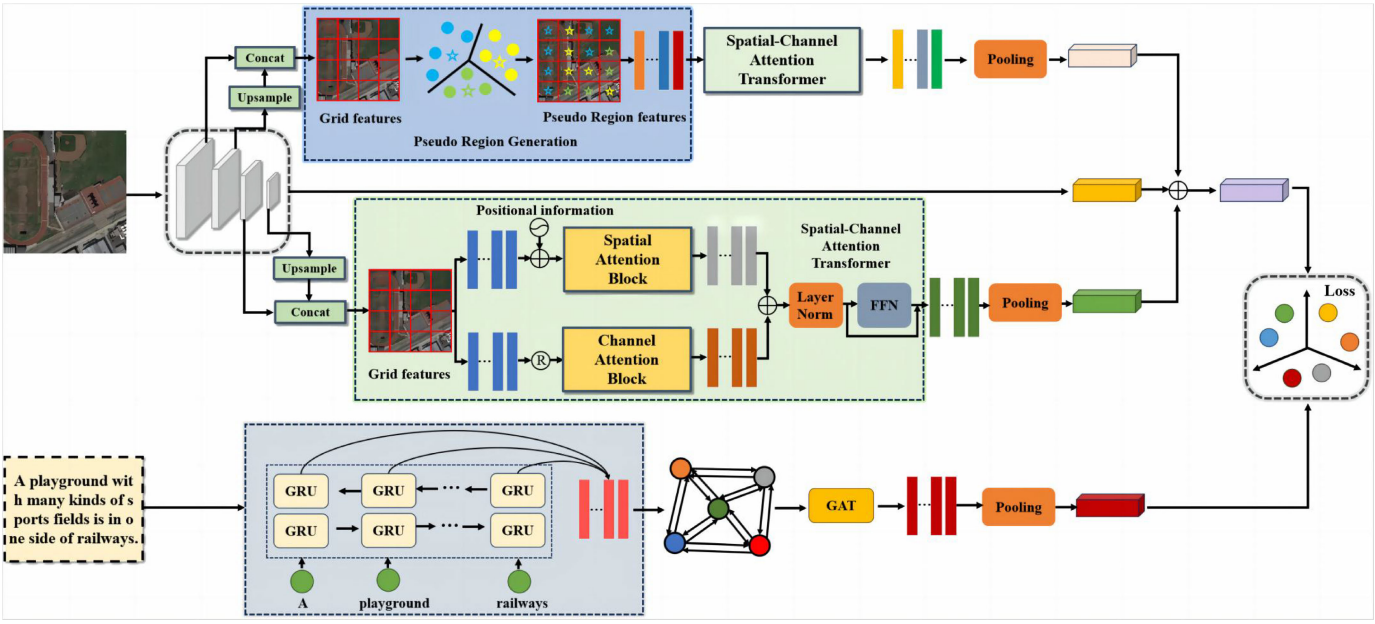


Fig. 2. Overview of our model SCAT-PRG. For the image part, we first utilize a pretrained ResNet-50 network to extract multilevel visual features. The PRG module is adopted to aggregate semantically similar low-level grid features to yield multiple pseudo regions. Then, we employ the proposed SCAT to model the dependency between features in spatial and channel domains. With multiple pooling modules and a simple addition operation, we can obtain the final global image representation. For the text part, we use Bi-GRU and GATs to learn the relation-enhanced word representations, followed by a pooling module to obtain the final global text representation. Finally, the whole model is optimized through a triplet ranking loss function during the end-to-end training. Ⓡ represents the reshape operation.

with the same shape as c_m. This design of introducing two cluster centers is to further enhance the adaptability of the PRG module. Finally, the learned $v^{\mathrm{p}} = \{v_1^{\mathrm{p}}, v_2^{\mathrm{p}}, \ldots, v_M^{\mathrm{p}}\}$ is defined as pseudo region features.

Specifically, our PRG module has two significant advantages. On the one hand, compared to detection-based methods, this module is more flexible and does not require any form of label supervision information. Its parameters can be updated end-to-end throughout the whole network using backpropagation. On the other hand, the number of pseudo regions obtained by clustering is much smaller than the number of low-level grid features (i.e., $M \ll N$), greatly reducing the computational cost involved in subsequent attention operations and making the model's inference faster.

C. Spatial–Channel Attention Transformer

After obtaining the high-level visual features v^{h} and a set of pseudo region features v^{p} generated from low-level visual features v^{l}, exploring the correlations between features to gain a comprehensive understanding of the rich semantic information in remote sensing images is another challenging task. Toward this end, we propose a novel SCAT module that mines fine-grained interactions between features in the spatial and channel domains to enhance the expression capability of visual features. Specifically, the SAB is utilized to capture spatial dependencies, highlighting important objects and regions in remote sensing images, enabling the model to know "where to look." On the other hand, the CAB is employed to capture channel dependencies, emphasizing semantically important feature dimensions, allowing the model to understand "what it is." By combining spatial and channel attention mechanisms,

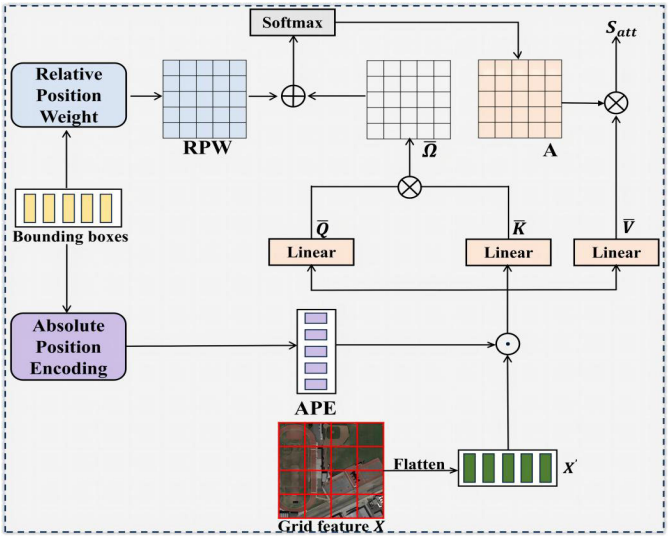


Fig. 3. Illustration of our proposed SAB. $\odot$ and $\otimes$, respectively, represent the element-wise product and matrix multiplication.

SCAT can effectively explore and leverage both spatial and semantic information to improve the performance of RSITR.

1) Spatial Attention Block: Due to the powerful modeling capabilities of the SA, using it as the SAB to capture spatial dependencies is entirely feasible. Specifically, given the input grid features $X \in \mathbb{R}^{W \times H \times C}$, we first need to flatten them in the spatial domain to obtain $X' \in \mathbb{R}^{N \times W}$, where $N = WH$. Then, the SA maps X' to corresponding query $Q \in \mathbb{R}^{N \times C}$, key $K \in \mathbb{R}^{N \times C}$, and value $V \in \mathbb{R}^{N \times C}$ through three linear embedding layers

$$Q = W_q X' + b_q, \quad K = W_k X' + b_k, \quad V = W_v X' + b_v \tag{7}$$

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where $W_{\{q,k,v\}}$ and $b_{\{q,k,v\}}$ are learnable parameters. Finally, the affinity between the ith and jth element can be computed as follows:

$$\Omega_{ij} = \frac{Q_i K_j^T}{\sqrt{d}} \tag{8}$$

where $\sqrt{d}$ is a scaling factor. However, the flattening operation inevitably leads to the loss of positional information. To make the above process sensitive to positions and capture spatial dependencies more accurately, we further introduce positional information in two ways: absolute position encoding and relative position weight. The final proposed SAB is illustrated in Fig. 3.

a) Absolute position encoding: Since grid can be considered a special case of region, we can also obtain the bounding box $B_i = \{x_i, y_i, w_i, h_i\}$ for arbitrary grid, where x_i, y_i, w_i, and h_i mean the center coordinates, width, and height. The PyTorch code provided in Fig. 4 elaborates on the detailed process of calculating grid bounding boxes. Given the B_i, we add two additional dimensions and normalize them, followed by a fully connected layer to obtain the absolute position encoding

$$p_i = \left(\frac{x_i}{w}, \frac{y_i}{h}, \frac{w_i}{w}, \frac{h_i}{h}, \frac{w_i}{h_i}, \frac{w_i h_i}{wh}\right) \tag{9}$$

$$\mathrm{APE}_i = W_p p_i + b_p \tag{10}$$

where w and h represent the width and height of the original image, and W_p and b_p are learnable parameters.

b) Relative position weight: We obtain the relative position weights between two grids by simulating the computation of relative position weights for regions

$$r_{ij} = \log\left(\frac{|x_i - x_j|}{w_i}, \frac{|y_i - y_j|}{h_i}, \frac{w_i}{w_j}, \frac{h_i}{h_j}\right) \tag{11}$$

$$\mathrm{RPW}_{ij} = \mathrm{ReLU}\left(w_{r2}(w_{r1} r_{ij} + b_{r1}) + b_{r2}\right) \tag{12}$$

where $w_{\{r_1,r_2\}}$ and $b_{\{r1,r2\}}$ are learnable parameters. RPW $\in \mathbb{R}^{N\times N}$ is the relative position weight. The nonnegativity of the rectified linear unit (ReLU) ensures that the model only considers relationships between grids with geometric correlations.

By integrating absolute position encoding and relative position weights into the traditional SA, the computation of our proposed SAB is as follows:

$$\bar{X} = X' \odot \mathrm{APE} \tag{13}$$

$$\bar{Q} = W_q \bar{X} + b_q, \quad \bar{K} = W_k \bar{X} + b_k, \quad \bar{V} = W_v \bar{X} + b_v \tag{14}$$

$$\bar{\Omega}_{ij} = \frac{\bar{Q}_i \bar{K}_j^T}{\sqrt{d}}, \quad A_{ij} = \frac{\exp(\bar{\Omega}_{ij} + \mathrm{RPW}_{ij})}{\sum_{k=1}^{N} \exp(\bar{\Omega}_{ik} + \mathrm{RPW}_{ik})} \tag{15}$$

$$S_{\mathrm{att}} = A\bar{V} \tag{16}$$

where S_{att} is the final spatial attention feature map and $\odot$ means the element-wise product. Similar to the traditional SA, our proposed SAB is also multiheaded and is omitted here.

2) Channel Attention Block: To effectively model the interaction between feature channels, using another SA as a CAB is theoretically feasible. However, since the computational complexity of SA is $O(N^2)$, where N is the length of the input sequence, it greatly increases the computational burden of the model when the input dimension C is high. In addition, SA may not fully consider the relationship between different channels, and the interaction between channels can also become very complex. To address this issue, we propose an efficient channel attention module that introduces the idea of 1-D convolution from ECA-Net [62] into nonlocal networks [63] and achieves higher performance with less computational cost.

```python
def get_relative_pos(x, bs, grid_size):
    x = x.view(1, -1, 1).expand(bs, -1, -1)
    return  x / grid_size

def get_grids_pos(bs, grid_size):
    x = torch.arange(0, grid_size).float().cuda()
    y = torch.arange(0, grid_size).float().cuda()
    px_min = x.view(-1, 1).expand(-1, grid_size).contiguous().view(-1)
    py_min = y.view(1, -1).expand(grid_size, -1).contiguous().view(-1)
    px_max = px_min + 1
    py_max = py_min + 1

    x_min = get_relative_pos(px_min, bs, grid_size)
    y_min = get_relative_pos(py_min, bs, grid_size)
    x_max = get_relative_pos(px_max, bs, grid_size)
    y_max = get_relative_pos(py_max, bs, grid_size)
    x = (x_min + x_max) * 0.5
    y = (y_min + y_max) * 0.5
    w = (x_max - x_min) + 1.
    h = (y_max - y_min) + 1.
    return x, y, w, h
```

Fig. 4. Detailed process of calculating grid bounding boxes. In the function "get_grids_pos," "bs" represents the batch size, and "grid_size" represents the width or height of the input grid features.

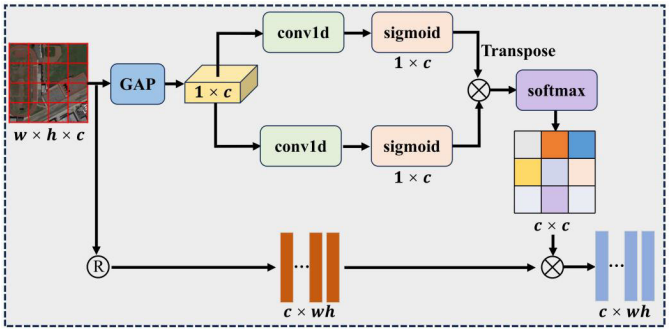


Fig. 5. Illustration of our proposed CAB. Ⓡ and ⊗, respectively, represent the reshape operation and matrix multiplication.

As shown in Fig. 5, given the input grid features $X \in \mathbb{R}^{W\times H\times C}$, we first employ global average pooling (GAP) to aggregate information in the spatial domain, which can avoid unnecessary parameters and calculations

$$F_c = \mathrm{GAP}(X)_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_{ij,c} \tag{17}$$

where $X_{ij,c}$ represents the pixel value of the cth channel in the ith row and jth column, while $F_c \in \mathbb{R}^1$ represents the aggregated information of the cth channel after pooling. $F \in \mathbb{R}^{1\times C}$.

Then, we use a 1-D convolution with kernel size k and a sigmoid function to obtain the corresponding query tensor $Q_c \in \mathbb{R}^{1\times C}$ and key tensor $K_c \in \mathbb{R}^{1\times C}$ from F. Here, we do not reduce the dimensionality of the features to prevent

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information loss

$$Q_c = \sigma(\text{Conv1d}(F)), \quad K_c = \sigma(\text{Conv1d}(F)) \tag{18}$$

where Conv1d and σ, respectively, denote the 1-D convolution with kernel size of k and the sigmoid function. Meanwhile, we directly flatten the original feature tensor X in the spatial domain to obtain the value tensor $V_c \in \mathbb{R}^{(W\times H)\times C}$. Next, we compute the outer product of Q_c and K_c and apply the softmax function on the channel dimension to obtain the channel attention map $A_c \in \mathbb{R}^{C\times C}$

$$A_c = \text{softmax}\left(K_c^T Q_c\right). \tag{19}$$

Finally, we perform matrix multiplication between the channel attention map A_c and value tensor V_c, and reshape the result back to $W \times H \times C$ to obtain the channel attention feature map C_{att}

$$C_{\text{att}} = A_c V_c. \tag{20}$$

Compared with other channel attention methods, our method has less computational complexity. This is because when calculating the channel attention map, we only need to perform an outer product of two $1 \times C$ vectors, instead of performing two $WH \times C$ matrix multiplications. Moreover, 1-D convolution can better capture the local dependencies between channels, and feature reduction is not performed, which maximally preserves the information of the channels and enhances the expressive power of the features.

3) Spatial–Channel Attention Transformer: As shown in Fig. 2, we employ the parallel connection of spatial attention and CABs to implement the proposed SCAT module, which is supported by experimental results. Therefore, the final output of SCAT can be obtained using the following equation:

$$Z = \text{FFN}(\text{LN}(\text{drop}(S_{\text{att}}) + \text{drop}(C_{\text{att}}) + X)) \tag{21}$$

where LN and drop(), respectively, refer to layer normalization and dropout operations. FFN is a two-layer feed-forward network to further enhance the representation ability of features.

Based on SCAT, we can update the high-level features v^{h} and pseudo region features v^{p} as follows:

$$\hat{v}^{\text{h}} = \text{FFN}\left(\text{LN}\left(\text{drop}\left(S_{\text{att}}^{\text{h}}\right) + \text{drop}\left(C_{\text{att}}^{\text{h}}\right) + v^{\text{h}}\right)\right) \tag{22}$$

$$\hat{v}^{\text{p}} = \text{FFN}\left(\text{LN}\left(\text{drop}\left(S_{\text{att}}^{\text{p}}\right) + \text{drop}\left(C_{\text{att}}^{\text{p}}\right) + v^{\text{p}}\right)\right) \tag{23}$$

where $S_{\text{att}}^{\{\text{h,p}\}}$ and $C_{\text{att}}^{\{\text{h,p}\}}$ represent the spatial attention features and channel attention features about v^{h} and v^{p}, respectively. Here, we only introduce positional information when calculating $\hat{v}^{\text{h}}$, as the generated pseudo regions v^{p} do not have explicit bounding boxes. By performing pooling operations on $\hat{v}^{\text{h}}$ and $\hat{v}^{\text{p}}$ and then fusing them with the global feature v^{g}, we can obtain the final global visual embedding

$$V' = \text{GPO}(\hat{v}^{\text{h}}) + \text{GPO}(\hat{v}^{\text{p}}) + v^{\text{g}} \tag{24}$$

where generalized pooling operator (GPO) is the pooling method proposed in [64] that has shown remarkable performance improvement in image–text retrieval tasks compared to traditional pooling strategies (i.e., mean-pooling and max-pooling). Therefore, we have chosen it as our pooling strategy.

D. Textual Representation

Given a sentence S with L words, we first encode each word w_i as a one-hot vector and use a learnable embedding layer to map these one-hot vectors to dense vectors of 300 dimensions u_i. Then, a Bi-GRU is used to encode the semantic context of the sentence from both forward and backward directions

$$\overrightarrow{h_i} = \overrightarrow{\text{GRU}}\left(u_i, \overrightarrow{h_{i-1}}\right), \quad \overleftarrow{h_i} = \overleftarrow{\text{GRU}}\left(u_i, \overleftarrow{h_{i+1}}\right) \tag{25}$$

where $\overrightarrow{h_i}$ and $\overleftarrow{h_i}$ denote the forward and backward hidden states, respectively. By averaging them, the representation of the word w_i is defined as $t_i = ((\overrightarrow{h_i} + \overleftarrow{h_i})/2)$. Similarly, we use a fully connected layer to map t_i into a d-dimensional common space. Finally, the sentence S can be represented as $T = \{t_i | i = 1, 2, \ldots, L, t_i \in \mathbb{R}^d\}$.

To further capture the high-level contextual relationships, we construct a fully connected graph $G = (T, E_T)$. All word features form the node set T, and the affinity matrix between words forms the edge set E_T. Utilizing graph attention networks (GATs) [65] to reason on this fully connected graph, the information of neighbor nodes can be efficiently aggregated. Specifically, we first compute the affinity coefficient between any two graph nodes

$$\mu_{ij} = a\left(W_q' t_i, W_k' t_j\right) \tag{26}$$

$$a\left(W_q' t_i, W_k' t_j\right) = \left(W_q' t_i \left(W_k' t_j\right)^T\right)/\sqrt{d} \tag{27}$$

$$e_{ij} = \text{softmax}(\mu_{ij}) = \frac{\exp(\mu_{ij})}{\sum_{j\in\mathbb{N}_i}\exp(\mu_{ij})} \tag{28}$$

where $W'_{\{q,k\}}$ is the learnable parameters, and $\mathbb{N}_i$ is the neighborhood of node i in the graph G. Since G is a fully connected graph, $\mathbb{N}_i$ contains all nodes in G. Using the attention coefficient, we can obtain the output features for node t_i

$$\bar{t}_i = \sum_{j\in\mathbb{N}_i} e_{ij} W_t t_j \tag{29}$$

where W_t is the weight matrix. To reduce the parameters and stabilize the training process, we perform the above process using multiple heads, which are omitted here. Finally, we employ the other pooling module GPO on context-aware word features $\bar{T} = \{\bar{t}_i | i = 1, 2, \ldots, L, \bar{t}_i \in \mathbb{R}^d\}$ to obtain the global textual embedding

$$T' = \text{GPO}(\bar{T}). \tag{30}$$

E. Object Function

We employ the mainstream triplet ranking loss in image–text retrieval task as the objective function to optimize the whole model, which aims to minimize the distance between anchor samples and their corresponding positive samples while maximizing the distance between anchor samples and negative samples. It is defined as follows:

$$L_r\left(V', T'\right) = \sum_{\hat{V}}^{'} \left[\lambda - S\left(V', T'\right) + S\left(\hat{V}', T'\right)\right]_+ + \sum_{\hat{T}'} \left[\lambda - S\left(V', T'\right) + S\left(V', \hat{T}'\right)\right]_+ \tag{31}$$

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where λ acts as a margin value, and $[x]_+ \equiv \max(x, 0)$. S is the similarity score between the remote sensing image and the text, and we choose cosine similarity in the experiment. $\hat{V}'$ and $\hat{T}'$ represent the negative samples corresponding to T' and V', respectively. In practice, we select all negative samples to participate in the calculation of the loss function. In order to speed up the training efficiency, we select the negative samples in each mini-batch instead of the whole training set.

IV. Experiments

A. Datasets and Evaluation Metrics

1) Datasets: We train and evaluate our proposed model on two widely used remote sensing image text retrieval datasets, RSICD [29] and RSITMD [16]. The number of images per scene category in the two datasets is shown in Table I, respectively. Each image in these datasets corresponds to five descriptive sentences. Specifically, RSICD is currently the largest RSITR dataset, comprising a total of 10 921 images, each with a size of 224 × 224, corresponding to 54 605 sentences. RSITMD is a more fine-grained dataset, consisting of a total of 4743 images, each with a size of 256 × 256 pixels, corresponding to 23 715 sentences. We follow previous data splitting methods [4], [7], [16], [20], using 80% of the samples in each dataset for training, 10% for validation, and the remaining 10% for testing.

2) Evaluation Metrics: We utilize recall at K (R@K) as the evaluation metrics to evaluate the performance of the model, which is most commonly used in image–text retrieval tasks. Specifically, R@K indicates the percentage of ground-truth matching in the top-K candidate results. Following previous work [7], [16], [20], K is set to 1, 5, and 10. In addition, we also select two other metrics, "mR" and "Rsum," that better reflect the overall performance of the model. "mR" is the average recall rate, and "Rsum" is the sum of the recall rates. The higher the R@K, "mR," and "Rsum" scores, the better the performance of the model.

B. Implementation Details

All experiments in the article are conducted on a single NVIDIA Tesla V100 GPU with the PyTorch platform. For images of different sizes, we uniformly resize them to 256 × 256 pixels and employ a series of data augmentation methods to enhance model robustness. We utilize the ResNet-50 network pretrained on the ImageNet dataset as the backbone network to extract multilevel visual features. This choice is made to ensure experimental consistency and prevent overfitting caused by excessive parameters in deep networks. The dimension of the joint embedding space is set to 512 (i.e., $d = 512$), and the number of pseudo regions (M) is set to 9. The margin value for the triplet loss function is 0.2, and the Adam optimizer is used to train the entire network. We train the model for 20 epochs using a batch size of 64 on both datasets. The initial learning rate is set to 0.0002 and decays tenfold at the halfway point of training. After each epoch, we validate the model on the validation set and select the snapshot with the highest Rsum score for testing. To reduce the impact of data partitioning on performance, for each set of experiments in the article, we use the same data partitioning method to construct five different sets of training and validation data and report their average results as the final experimental results.

C. Comparisons With Other Methods

In this section, we compare our proposed method with the following cross-modal image–text retrieval methods: VSE++ [66], SCAN [22], CAMP [23], MTFN [67], CAMERA [68], LW-MCR [4], AMFMN [16], KCR [17], CMFM-Net [19], SSJDN [21], HyperMatch [25], GaLR [20], and SWAN [7]. For the first five methods, as they are designed for natural image–text datasets, we modify their source code to adapt to remote sensing image–text datasets, fine-tune specific parameters, and conduct cross-validation to ensure fair comparison results. For the remaining methods specifically designed for RSITR, we directly report the experimental results documented in their original papers. Table II presents the comparison results of our method and other comparison methods on the RSICD and RSITMD datasets, with the best results indicated in bold. From this table, we can draw the following conclusions.

1) Our method achieves nearly optimal performance on all evaluation metrics for both datasets, surpassing current mainstream RSITR methods by a large margin. Specifically, on the RSICD dataset, our method achieves R@1 scores of 10.88% and 8.06% for text retrieval and image retrieval, which represent significant improvements compared to the previous best method SWAN [7], with increases of 3.47% and 2.50%, respectively. This is primarily attributed to our proposed PRG module and SCAT, which enable our model to accurately perceive local objects in the image and also have a comprehensive understanding of the semantic content. In addition, compared to GaLR [20], our method not only outperforms it in terms of performance (e.g., our model shows a 6.15% improvement in mR) but also eliminates the need for an additional object detector, making our model more convenient in both the training and testing processes. Similarly, on the RSITMD dataset, our method achieves the best performance on most evaluation metrics, consistent with the results on the RSICD dataset. Overall, these experimental results fully demonstrate the effectiveness and superiority of our proposed method.
2) Compared to approaches specifically designed for remote sensing images, traditional image–text retrieval methods (e.g., VSE++ [66], SCAN [22], and MTFN [67]) designed for natural images tend to exhibit relatively poorer performance on both datasets. This proves that directly transferring traditional image–text retrieval models to the RSITR domain through parameter fine-tuning is not feasible. Our approach, along with other specialized methods, pays closer attention to the specific challenges that exist in the remote sensing domain. By addressing these issues, these methods can learn more distinctive feature representations, thereby achieving superior retrieval performance.
3) Different backbones yield different results. Specifically, using ResNet-50 as the backbone achieves the best

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TABLE I

SCENE CATEGORIES AND THEIR CORRESPONDING NUMBERS IN THE RSICD AND RSITMD DATASET

Category	Number (RSICD,RSITMD)	Category	Number (RSICD,RSITMD)	Category	Number (RSICD,RSITMD)	Category	Number (RSICD,RSITMD)
Airport	420, 140	Dense Residential	410, 157	Park	350, 160	Railway Station	260, 112
Bare Land	310, 58	Desert	300, 126	School	300, 165	Resort	290, 166
Baseball Field	276, 105	Farmland	370, 177	Square	330, 196	River	410, 188
Beach	400, 154	Forest	250, 116	Parking	390, 143	Sparse Residential	300, 157
Bridge	459, 155	Industrial	390, 207	Playground	1031, 191	Storage Tanks	396, 201
Center	260, 146	Meadow	280, 127	Pond	420, 184	Stadium	290, 201
Church	240, 136	Medium Residential	290, 130	Viaduct	420, 188	Boat	0, 55
Commercial	350, 148	Mountain	340, 109	Port	389, 178	Intersection	0, 67

TABLE II

BIDIRECTIONAL RETRIEVAL RESULTS ON THE RSICD AND RSITMD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. † AND ‡ REPRESENT THE USE OF RESNET-50 AND RESNET-101 AS BACKBONE TO EXTRACT VISUAL FEATURES, AND RESNET-18 OTHERWISE

Method	RSICD								RSITMD							
	Image-to-Text			Text-to-Image			mR	Rsum	Image-to-Text			Text-to-Image			mR	Rsum
	R@1	R@5	R@10	R@1	R@5	R@10			R@1	R@5	R@10	R@1	R@5	R@10		
Traditional Image-Text Retrieval																
VSE++ [66]	3.38	9.51	17.46	2.82	11.32	18.10	10.43	62.59	10.38	27.65	39.60	7.79	24.87	38.67	24.82	148.93
SCAN t2i [22]	4.39	10.90	17.64	3.91	16.20	26.49	13.25	79.53	10.18	28.53	38.49	10.10	28.98	43.53	26.64	159.81
SCAN i2t [22]	5.85	12.89	19.84	3.71	16.40	26.73	14.24	85.42	11.06	25.88	39.38	9.82	29.38	42.12	26.27	157.64
CAMP-triplet [23]	5.12	12.89	21.12	4.15	15.23	27.81	14.39	86.32	11.73	26.99	38.05	8.27	27.79	44.34	26.20	157.17
CAMP-bce [23]	4.20	10.24	15.45	2.72	12.76	22.89	11.38	68.26	9.07	23.01	33.19	5.22	23.32	38.36	22.03	132.17
MTFN [67]	5.02	12.52	19.74	4.90	17.17	29.49	14.81	88.84	10.40	27.65	36.28	9.96	31.37	45.84	26.92	161.50
CAMERA [68]	4.57	13.08	21.77	4.00	15.93	26.97	14.39	86.32	8.33	21.83	33.11	7.52	26.19	40.72	22.95	137.70
Remote Sensing Image-text Retrieval																
LW-MCR [4]	3.29	12.52	19.93	4.66	17.51	30.02	14.66	87.93	10.18	28.98	39.82	7.79	30.18	49.78	27.79	166.73
AMFMN-soft [16]	5.05	14.53	21.57	5.05	19.74	31.04	16.16	96.98	11.06	25.88	39.82	9.82	33.94	51.90	28.74	172.42
AMFMN-fusion [16]	5.39	15.08	23.40	4.90	18.28	31.44	16.42	98.49	11.06	29.20	38.72	9.96	34.03	52.96	29.32	175.93
AMFMN-sim [16]	5.21	14.72	21.57	4.08	17.00	30.60	15.53	93.18	10.63	24.78	41.81	11.51	34.69	54.87	29.72	178.29
KCR‡ [17]	4.76	18.59	27.20	5.84	22.31	36.12	19.14	114.82	-	-	-	-	-	-	-	-
CMFM-Net [19]	5.40	18.66	28.55	5.31	18.57	30.03	17.75	106.52	10.84	28.76	40.04	10.00	32.83	47.21	28.28	169.68
SSJDN [21]	6.50	19.70	30.10	4.90	20.20	36.50	19.65	117.90	12.20	29.40	44.20	10.80	**42.20**	**68.90**	34.62	207.70
HyperMatch [25]	7.14	20.04	31.02	6.08	20.37	33.82	19.75	118.47	11.73	28.10	38.05	9.16	32.31	46.64	27.67	165.99
GaLR [20]	6.59	19.85	31.04	4.69	19.48	32.13	18.96	113.78	14.82	31.64	42.48	11.15	36.68	51.68	31.41	188.45
SWAN† [7]	7.41	20.13	30.86	5.56	22.26	37.41	20.61	123.63	13.35	32.15	46.90	11.24	40.40	60.60	34.11	204.64
SCAT-PRG(ResNet-18)	10.13 ±0.21	25.72 ±0.37	38.24 ±0.58	7.76 ±0.20	25.20 ±0.42	38.69 ±0.31	24.29 ±0.28	145.74 ±0.72	**16.82** ±0.44	32.73 ±0.36	48.12 ±0.57	12.11 ±0.31	42.15 ±0.26	61.97 ±0.88	**35.65** ±0.59	**213.90** ±1.03
SCAT-PRG(ResNet-50)	**10.88** ±0.38	**26.32** ±0.42	**39.66** ±0.64	**8.06** ±0.23	**25.43** ±0.52	**40.31** ±0.54	**25.11** ±0.29	**150.66** ±0.51	16.50 ±0.47	**33.25** ±0.26	**48.49** ±0.43	**12.55** ±0.28	41.86 ±0.22	61.07 ±1.05	35.62 ±0.42	213.72 ±1.21
SCAT-PRG(ResNet-101)	10.78 ±0.49	25.92 ±0.39	38.66 ±0.57	7.87 ±0.29	25.31 ±0.48	39.78 ±0.51	24.72 ±0.39	148.32 ±0.68	16.62 ±0.49	32.57 ±0.32	48.32 ±0.54	11.74 ±0.30	41.50 ±0.31	60.87 ±1.15	35.27 ±0.57	211.62 ±1.36

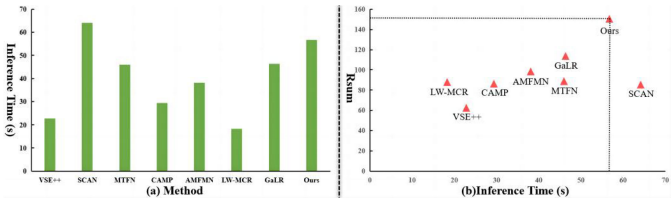


Fig. 6. (a) Inference time for different methods. (b) Accuracy-speed tradeoff on the RSICD dataset. "s" denotes seconds.

results, with an improvement of 4.92% and 2.34% in Rsum on the RSICD dataset compared to ResNet-18 and ResNet-101, respectively. While there may be a slight decrease in the model's performance on the RSITMD dataset, this is primarily due to the limited amount of data available, as the entire dataset consists of only 4743 images. This finding aligns with observations made in previous papers [61], indicating that with limited dataset size, deeper ResNet networks do not necessarily further enhance visual representations.

Furthermore, we also compare the inference time of the proposed model with other models because retrieval efficiency remains a crucial factor in evaluating model performance. Specifically, inference time refers to the time to compute similarity scores for all image–text pairs. Seven representative algorithms are used for comparison under the same experimental environment on the RSICD dataset. The comparison results are shown in Fig. 6, where the left subgraph [see Fig. 6(a)] represents the time comparison of different methods, and the right subgraph [see Fig. 6(b)] illustrates the accuracy-speed tradeoff. The closer a point is to the top-left corner, the better the overall performance of the model. By observing these results, we find that the inference time of our model is only slightly faster than SCAN, which employs a complex local matching strategy, and slower than the other compared models. The fastest model is LW-MCR. The primary reason for this phenomenon is that our SCAT module consumes a significant amount of time when exploring the interactions between features in the channel and spatial domains. However, this inference time still falls within a reasonable range, being only 10.38 s slower than GaLR on the RSICD dataset. Moreover, the time cost has brought significant performance improvements to our model, as shown in Fig. 6(b), indicating that the additional time overhead is worthwhile.

D. Ablation Study

In this section, we conduct detailed ablation experiments on the RSICD dataset using ResNet-50 as the visual backbone, in order to comprehensively analyze our proposed model.

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODULE ABLATION FOR OUR PROPOSED MODEL ON THE RSICD DATASET. "✓" DENOTES RETAINING IT, OTHERWISE REMOVING IT

Model	Visual Feature			PRG	SCAT	GAT	Image-to-Text			Text-to-Image			mR	Rsum
	v^l	v^h	v^g				R@1	R@5	R@10	R@1	R@5	R@10		
m1			✓				7.88	21.30	33.83	6.16	21.20	34.85	20.87	125.22
m2	✓		✓				8.09	22.60	34.09	6.20	22.47	36.29	21.64	129.84
m3		✓	✓				8.22	22.65	34.28	6.31	22.50	36.54	21.74	130.50
m4	✓	✓	✓				8.35	23.14	35.41	6.42	22.84	37.10	22.21	133.26
m5	✓	✓	✓			✓	8.85	23.98	36.04	7.02	23.86	38.19	22.99	137.94
m6	✓	✓	✓	✓			9.04	23.75	36.83	6.65	22.87	36.58	22.62	135.72
m7	✓	✓	✓		✓		9.34	24.25	37.18	6.84	22.97	37.66	23.04	138.24
m8	✓	✓	✓	✓		✓	9.36	23.88	36.28	7.10	23.72	38.14	23.08	138.48
m9	✓	✓	✓		✓	✓	10.39	25.71	39.33	7.72	24.74	39.17	24.51	147.06
Full Model	✓	✓	✓	✓	✓	✓	**10.88**	**26.32**	**39.66**	**8.06**	**25.43**	**40.31**	**25.11**	**150.66**

1) Performance Analysis of Different Components: To investigate the impact of different key components, we designed ten different ablation experiments, shown as follows.

1) *m1:* Using only global visual features v^{g} extracted by CNN and textual features T extracted by Bi-GRU, which is our base model.
2) *m2 and m3:* Adding low-level visual feature v^{l} and high-level visual feature v^{h} separately on the basis of $m1$.
3) *m4:* Adding multilevel visual features v^{h} and v^{l} on the basis of $m1$.
4) *m5:* Using textual features T' extracted by GATs and visual features same as $m4$.
5) *m6 and m7:* Adding the proposed PRG module and SCAT module separately on the basis of $m4$.
6) *m8 and m9:* Adding the proposed PRG module and SCAT module separately on the basis of $m5$.
7) *m10:* This is our final proposed model.

Table III provides a detailed description of the performance of different ablation versions of our model. We find that the base model has the lowest performance, with R@1 scores of 7.88% and 6.16% for sentence retrieval and image retrieval, respectively. The experimental results for m2 and m3 indicate that whether it is increasing low-level or high-level grid features, the model's performance improves to some extent. Moreover, combining both types of features results in a more significant performance boost, as demonstrated by m4, which shows a 1.34% improvement in mR compared to the base model, highlighting the effectiveness of multilevel visual features. Comparing m5 and m4, which have the same visual features, it is evident that when using a GAT to model relationships between words in sentences, the model's performance improves correspondingly. Furthermore, we observe that for all experiments using GAT, the model's performance in the text-to-image retrieval direction is noticeably better than experiments without GAT. This is because when using GAT for inference on the fully connected textual graph, each word can aggregate information from adjacent words to enhance feature representations. In addition, when equipped with the PRG module, the model's performance further improves. For example, comparing m5–m8, the R@1 scores for sentence retrieval increases by 0.51%. By comparing m4 and m6, we can draw similar conclusions. Comparing m4 with m7 or m5 with m9, it can be observed that when using the SCAT module, the model achieves better performance, with significant improvements. This is because by modeling interactions between features in both the spatial and channel domains, the model can focus more on semantically rich regions or channels, helping to extract more discriminative and informative features. Finally, with the combination of all key components, our model achieves the best retrieval performance. The above ablation experiments provide strong evidence for the effectiveness of our model, where each key module contributes differently to the overall retrieval performance.

2) Spatial–Channel Attention Transformer: To effectively model the correlation of features in both spatial and channel domains, we have carefully designed the SCAT module. Therefore, in this section, we conduct several ablation experiments to further evaluate the impact of each attention block on model performance. Specifically, we begin by removing the entire SCAT module, CAB, and SAB from the overall model, denoted as "w/o SCAT," "w/o CAB," and "w/o SAB," respectively. Next, we successively remove the absolute position encoding APE, the relative position weight RPW, and both of them together, defined as "w/o SAB_APE," "w/o SAB_RPW," and "w/o SAB_AR." Finally, we replace CAB with SA to model the dependencies between features in the channel domain, referred to as "w SAB + SA." In addition, we also calculate the parameters of different architectures to demonstrate the effectiveness of SCAT.

The comparison results are shown in Table IV. We can see that the retrieval performance of the "w/o SCAT" model severely declined, with R@1 decreasing by 1.52% and 0.96% for sentence retrieval and image retrieval, respectively. This indicates that SCAT can capture richer semantic information in remote sensing images by modeling the intrinsic correlation between features, thereby improving the performance of the model. Lines 3 and 4 demonstrate the effectiveness of

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TABLE IV

ABLATION STUDY OF THE SCAT USED IN THE MODEL ON THE RSICD DATASET. "SA" AND "CA," RESPECTIVELY, REFER TO SPATIAL ATTENTION AND CHANNEL ATTENTION

Model	Image-to-Text			Text-to-Image			mR	Rsum	Params(M)
	R@1	R@5	R@10	R@1	R@5	R@10			
Full Model	**10.88**	**26.32**	**39.66**	**8.06**	**25.43**	**40.31**	**25.11**	**150.66**	40.7629
w/o SCAT	9.36	23.88	36.28	7.10	23.72	38.14	23.08	138.48	34.4521
w/o CAB	9.84	24.25	37.47	7.52	25.08	38.70	23.81	142.86	40.7627
w/o SAB	9.78	25.22	37.86	7.56	24.38	38.42	23.87	143.22	38.6555
w/o SAB_APE	10.25	25.07	38.33	7.58	25.32	38.47	24.17	145.02	40.7594
w/o SAB_RPW	10.34	25.42	38.24	7.88	25.06	38.44	24.23	145.38	40.7624
w/o SAB_AR	9.94	24.82	38.24	7.69	24.90	38.65	24.04	144.24	40.7588
w SAB+SA	9.80	24.39	37.31	7.42	24.28	38.76	23.66	141.96	**41.3096**

each attention module. For example, compared to the "w/o SCAT" model, the addition of CAB ("w/o SAB") results in an improvement of 0.42% and 0.46% in R@1 for the two retrieval directions, respectively. We also observe that the CAB performs better than the SAB when a single attention module is adopted. In addition, we can observe that these two attention modules are complementary. By simultaneously mining the correlation between features in both spatial and channel domains, we can not only enable the model to know "where to look," but also "what it is," thus generating more discriminative and robust feature representations. Lines 5–7 demonstrate that introducing positional information in the SAB can effectively mitigate the information loss caused by the direct flattening operation and accelerate spatial relation modeling. In addition, we also find that both of the proposed ways to introduce positional information are effective. Line 8 further demonstrates the effectiveness of our proposed CAB. When using SA instead of the proposed CAB, the R@1 for both retrieval directions decreased by 1.08% and 0.64%, respectively.

From Table IV, we can observe that our proposed SCAT module, while adding 6.3108M parameters, results in a significant improvement of 12.18% in terms of Rsum. Importantly, our CAB, with only 0.0002M parameters, leads to a substantial performance boost. When using SA as a replacement for CAB, the parameter increases by 0.5469M, but the performance, with respect to Rsum, decreases by 8.7%. This implies that our CAB is able to model the dependencies between features on the channel domain in a more efficient and effective way.

3) More Combinations of Attention Modules: As the roles of the two attention modules are different, with spatial attention modeling spatial-wise dependencies and channel attention capturing channel-wise dependencies, their different combinations can have a significant impact on model performance. In this section, we construct three different ways to combine the SAB and the CAB. Specifically, the first method is to employ the SAB on the original features initially, followed by the CAB, i.e., "Spatial + Channel." The second method is to adopt the CAB on the original features initially, followed by the SAB, i.e., "Channel + Spatial." Both of them are serial combinations. The third method is parallel combination, i.e., "Spatial & Channel," where the two attention blocks are applied to the original features simultaneously, and the outputs of the two modules are further fused.

Table V shows the experimental results for different combinations of the two attention blocks. We can observe that

TABLE V

ABLATION STUDY ON DIFFERENT COMBINATIONS OF SPATIAL ATTENTION AND CHANNEL ATTENTION ON THE RSICD DATASET. "+" AND "&," RESPECTIVELY, REPRESENT THE SERIAL AND PARALLEL USE OF THESE TWO ATTENTION MODULES

Model	Image-to-Text			Text-to-Image			mR	Rsum
	R@1	R@5	R@10	R@1	R@5	R@10		
Spatial+Channel	10.28	25.96	39.46	7.68	24.34	39.34	24.51	147.06
Channel+Spatial	10.72	26.24	39.52	7.78	24.71	39.53	24.75	148.50
Spatial & Channel	**10.88**	**26.32**	**39.66**	**8.06**	**25.43**	**40.31**	**25.11**	**150.66**

TABLE VI

PERFORMANCE COMPARISON OF THE PRG MODULE WITH DIFFERENT CLUSTERING NUMBERS ON THE RSICD DATASET

Model	Image-to-Text			Text-to-Image			mR	Rsum
	R@1	R@5	R@10	R@1	R@5	R@10		
M=1	9.72	24.74	37.22	7.26	24.26	38.70	23.65	141.90
M=4	10.10	24.98	38.28	7.61	24.91	39.08	24.16	144.96
M=9	**10.88**	**26.32**	**39.66**	**8.06**	**25.43**	**40.31**	**25.11**	**150.66**
M=16	10.56	25.62	38.52	7.76	24.48	39.16	24.35	146.10
M=25	10.35	25.46	39.24	7.69	25.15	39.47	24.46	147.36

TABLE VII

PERFORMANCE COMPARISON OF THE PRG MODULE AT DIFFERENT POSITIONS ON THE RSICD DATASET

Model	Image-to-Text			Text-to-Image			mR	Rsum
	R@1	R@5	R@10	R@1	R@5	R@10		
PRG_$\{v^l\}$	**10.88**	**26.32**	**39.66**	**8.06**	**25.43**	**40.31**	**25.11**	**150.66**
PRG_$\{v^h\}$	8.05	21.85	33.40	6.45	22.47	36.12	21.39	128.34
PRG_$\{v^l, v^h\}$	8.62	22.53	34.52	6.74	22.71	36.34	21.91	131.46

the parallel combination of the two attention blocks performs better than the serial combination. For example, compared to "Spatial + Channel," "Spatial & Channel" improves R@1 by 0.6% and 0.38% in sentence retrieval and image retrieval, respectively. In addition, in most evaluation metrics, "Channel + Spatial" works better than "Spatial + Channel." Furthermore, when compared with the methods with using solely one attention module, all combination methods show better performance. From these results, we can draw two conclusions. On the one hand, it is crucial to simultaneously use both attention modules for fully mining information in images. On the other hand, exploring the optimal combination of the two attention modules can further improve model performance.

4) PRG Module: In our model, we propose a PRG module to aggregate grid features and obtain a fine-grained perception of local objects in images. There are two key factors in the PRG module: the number of generated pseudo regions and the location where the module should be applied. Therefore, we conducted two sets of ablation experiments to evaluate the effects of quantity and location on model performance, respectively, in order to gain a deeper understanding of the PRG module.

To determine how many pseudo regions the model needs to learn, we set the number of clusters from 1 to 25 and only took perfect squares. This is partly because exhaustive testing of all possible values of M is time-consuming and also for ease of subsequent attention operations. The experimental results are shown in Table VI. It can be seen that the performance of our model first increases and then decreases, with the increasing of

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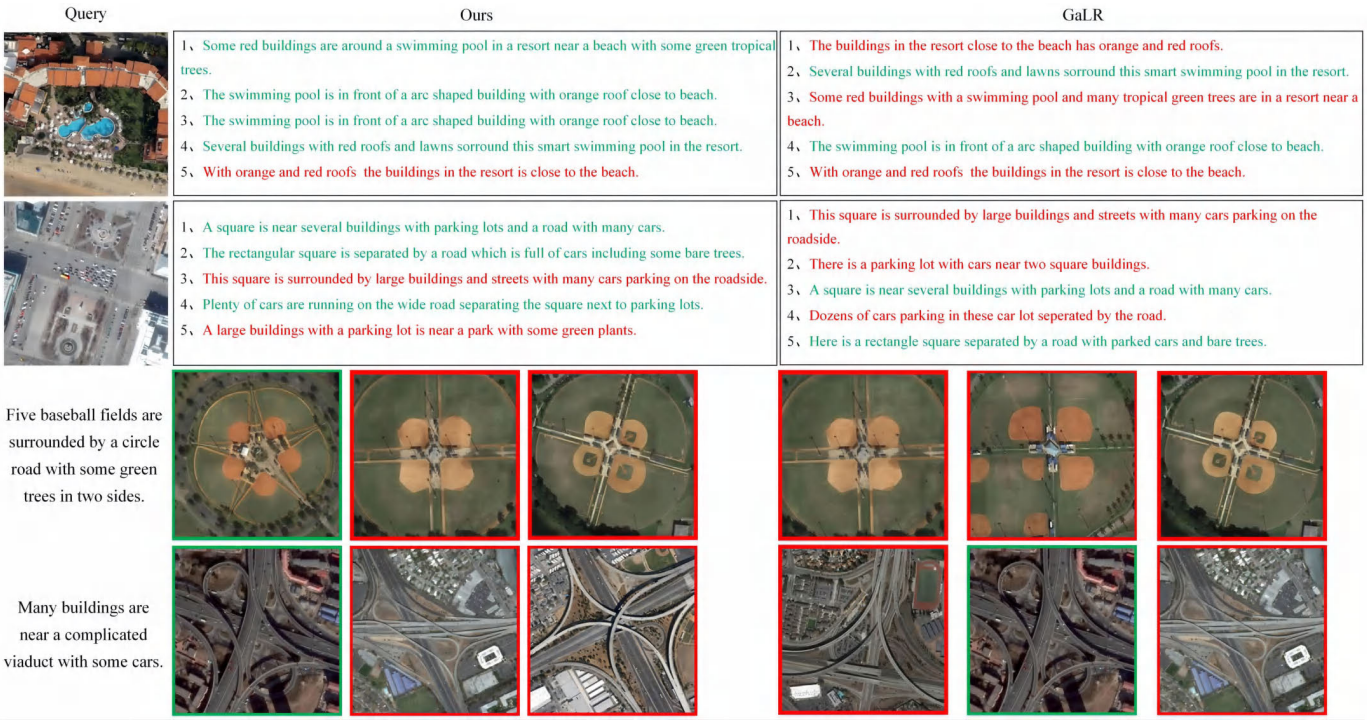


Fig. 7. Qualitative retrieval results of our model and GaLR [20] on the RSICD dataset. For each query, the ground-truth results are marked in green, while incorrect results are marked in red.

the number of generated pseudo regions. The observed results make sense because if the number of pseudo regions is too few, low-level grid features v^{l} may not be well aggregated, resulting in lower performance. Conversely, if there are too many pseudo regions, the model may generate some pseudo regions that have no semantic meaning, i.e., noise regions, which will prevent the model from perceiving distinguishable pseudo regions and ultimately harm its performance. Therefore, we set the number of generated pseudo regions to 9 in the experiment, which results in the best performance on all evaluation metrics.

Furthermore, to determine where the PRG module should be applied, we constructed three different models. The first two models apply the PRG module separately on low-level visual feature v^{l} and high-level visual feature v^{h}. The third model simultaneously applies the module on both v^{l} and v^{h}. Table VII shows the performance comparison results for the different models. As we can see, clustering on low-level visual features yields better results. This is because low-level visual features often contain more detailed information, such as edges, textures, and colors, which help to distinguish different local objects in the image and bring them together to discover relevant patterns and features. In contrast, high-level visual features usually have stronger semantic expression but may lack local information. Therefore, applying clustering algorithms on high-level features may be more susceptible to noise and interference, making it difficult to obtain accurate retrieval results.

E. Qualitative Results and Visualization Study

1) Qualitative Results: To further validate the effectiveness of the proposed model, we present some visualization results of sentence retrieval and image retrieval. In addition, we also showcase the bidirectional retrieval results of the GaLR [20] model to more intuitively reflect the advantages of our model. The visualization results are shown in Fig. 7. Specifically, for sentence retrieval, we list the top-5 retrieved sentences based on similarity scores calculated by different models. From the results, we observe that our model exhibits superior retrieval results compared to GaLR. Thanks to our proposed PRG module and SCAT module, the final visual representation can focus on the objects, their specific attributes, and the relationships between objects. For example, for the first query image that contains irregularly shaped objects like "swimming pool" and "buildings," the top-ranked sentences retrieved by our model accurately include them. In contrast, the GaLR model, which relies on an object detector, cannot accurately perceive these objects. Furthermore, even the incorrect sentences retrieved by our model also describe the content of the image.

For image retrieval, we have visualized the top three images retrieved by different models. Our model also successfully retrieves target images in the top-ranked positions, while other models do not retrieve them. In the first example, our model accurately captures the key discriminative concept "Five" from the sentence, leading to the correct retrieval result. In the second example, the key target word "car" in the sentence becomes crucial for our model to distinguish between these three similar images. Overall, these visualization results demonstrate the feasibility of our proposed model in the task of RSITR.

2) Visualization Study: To better understand the model's behavior, we use sentence retrieval as an example and visualize the similarity between query images and keywords in the sentences, explaining how our model retrieves the correct results. Specifically, we employ cosine distance to calculate

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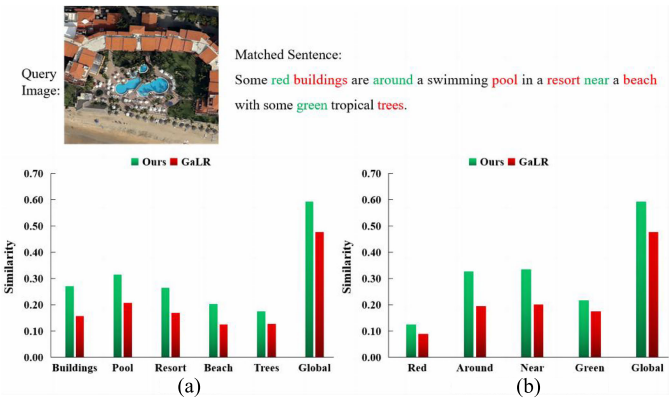


Fig. 8. Visualization of correlations between words and images. Higher scores are more relevant. For ease of understanding, some keywords in the sentences are highlighted. (a) Objects. (b) Attributes and relations.

the similarity between features, and the results are shown in Fig. 8. In Fig. 8, we compare and discuss the results in two scenarios to validate the advantages of the two key components in our model. In Fig. 8(a), we focus on the similarity between nouns in the sentence and the image, while in Fig. 8(b), we concentrate on the similarity between adjectives and prepositions in the sentence and the image. Similarly, we select GaLR [20] as the comparative model.

From Fig. 8, we can observe that for some keywords, the similarity scores generated by our model are significantly higher than those generated by GaLR. For example, for the word "pool" in Fig. 8(a), the similarity score generated by our model with the query image is 0.3153, while that generated by the comparison model is 0.2082, and the difference in similarity is 0.1071. For other nouns as well, there are indeed differences in similarity scores. This fully demonstrates the effectiveness of our proposed PRG module, which can obtain fine-grained perception of local objects in the image by clustering the semantically related grid features. Similarly, for the adjectives and prepositions (i.e., "around" and "green") in Fig. 8(b), we can draw similar conclusions. This means that our SCAT module can capture a more comprehensive understanding of the image content by modeling the correlation between features in the spatial and channel domains. In the end, the cumulative differences in similarity between these local objects and key attributes result in our model calculating a higher overall similarity than what GaLR calculates. Overall, these results fully prove that our model not only accurately perceives local objects in the image but also effectively models the relationships between objects, thereby improving the performance of RSITR.

V. Conclusion

In this article, we propose SCAT-PRG, a novel end-to-end framework for RSITR. Specifically, we first introduce a PRG module that generates multiple pseudo regions by performing clustering algorithms on low-level grid features, enabling fine-grained perception of objects in the image. Then, we carefully design a SCAT module that explores the correlations between features in both spatial and channel domains, leading to a more comprehensive understanding of the image content. Extensive experiments on two benchmark datasets, i.e., RSICD and RSITMD, demonstrate the effectiveness of the proposed method SCAT-PRG. In the future, we will investigate more efficient methods for exploring interactions between modalities and try to construct a larger and more fine-grained dataset to further advance RSITR.

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